Answers and talk notes

Describe Data Using the Mean

What your student is learning

Tonight we work the mean backwards: we know the average we WANT, and we find the score still needed to get there.

Answers

Ask your student to explain before you look.

1. **A.** 30. Target total = mean × count = 10 × 3 = 30.
2. **B.** 15. 60 − 45 = 15 more.
3. **C.** 8. Target total = 6 × 4 = 24. So far 4 + 7 + 5 = 16. 24 − 16 = 8.
4. **D.** 20. Target total = 16 × 5 = 80. So far 12 + 18 + 14 + 16 = 60. 80 − 60 = 20.
5. **B.** 14. Target total = 12 × 5 = 60. 10 + 14 + 9 + 13 = 46. 60 − 46 = 14.

Talk about it

How does a target mean become a target total?

Home mission

Pretend your family wants a mean of 6 glasses of water a day for 4 days. Days 1 to 3 were 5, 7 and 4. Ask your student how many glasses day 4 needs.

When they are stuck, ask

- If the mean is a fair share, what would the total have to be?
- How many values will there be when we are done — including the new one?
- Put your answer back in. Does the mean come out right?

Try not to say

- Just take the average of the scores he already has.
- Multiply by the number of tests he has taken so far.
- The missing score has to be close to the target, so guess.